

Lesson plan:

Name of the Programme:	B. Tech (H) CSE
Title of the Course:	Calculus and Laplace Transforms
Course code	CS2806
Semester:	III
Names of the Course Instructor(s)	Rammohan
	Hongray
	Ankita
	Sasikala
	Akhilesh
	Sahana (Guest Faculty)

Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
2	1+1+0	15L+15T

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
Session-1	Module-1	Functions – definitions, types, domain & range	Chalk & talk + Concept mapping	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-1
Session-2		Graphs – transformations, inverse, piecewise	Graph plotting using worksheets	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-1
Session-3		Limit concepts – basic and algebraic methods	Think-pair-share + problem-solving	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-1
Session-4		Continuity and types of discontinuities	Lecture + graphical interpretation using Desmos	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-1
Session-5	Module-2	Derivatives – definition and basic rules	Derivation using first principles + exercises	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-3
Session-6		Chain rule and implicit differentiation	Guided examples on board	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-3

Session-7		Taylor and Maclaurin series	Derivation of series + written exercises	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-11
Session-8		Linearization and approximation	Graphical visualization + concept Q&A	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-11
Session-9		Rate of change – physical and geometrical idea	Problem-solving + real-world interpretation	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-2
Session-10		Applications: increasing/decreasing, velocity	Group work with solved problems	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-4
Session-11		First derivative test, critical points	Case studies of functions + test application	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-4
Session-12		Second derivative test, curve sketching	Curve sketching with sign chart and intervals	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-4
Session-13	Module-3	Introduction to integration & antiderivatives	Chalk talk + sample computations	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-5
Session-14		Definite integrals and Fundamental Theorem	Exercises + flow diagrams	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-5
Session-15		Integration by substitution	Guided practice problems	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-5
Session-16		Integration by parts and review	Board work + peer correction	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-5
Session-17		Area under curves – single function	Definite integration problems	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-5
Session-18		Area between curves	Visual problems + comparative analysis	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-5

Session-19	Module-4	Partial derivatives – basic definitions	Multivariable function analysis	https://home.iitm.ac.in/asingh/papers/classnotes-ma1101.pdf ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-20		Tangent planes and level curves	Worksheets + 3D graphs	https://home.iitm.ac.in/asingh/papers/classnotes-ma1101.pdf ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-21		Multivariable Chain Rule	Concept building with chain maps	https://home.iitm.ac.in/asingh/papers/classnotes-ma1101.pdf ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-22		Total derivative and higher derivatives	Hands-on gradient computation	https://home.iitm.ac.in/asingh/papers/classnotes-ma1101.pdf ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-23		Gradient & critical points (2-variable)	Derivation and classification via examples	https://home.iitm.ac.in/asingh/papers/classnotes-ma1101.pdf ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-24	Module-5	Hessian, maxima/minima & saddle points	Curve tracing in space + problem solving	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-25		Vector functions & parametric curves	Curve tracing in space + problem solving	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-13

Session-26		Derivatives of vector functions	Chalk and talk + exercises	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-13
Session-27		Gradient and directional derivative	Concept maps + computation on scalar fields	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-13
Session-28		Tangent planes and normal vectors	Vector geometry sketches + formula-based problems	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-13
Session-29		Introduction to gradient descent	Iterative examples with fixed step sizes	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14
Session-30		Mathematical convergence using gradient ideas	Step-by-step gradient iterations	ISBN-13: 978-0-321-88407-7 Thomas calculus Chapter-14

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
- CIE-1: IA1 (Pen & Paper Theory): (CO1 - 10M; CO2 - 10M)	20
- CIE-2: Midterm (Pen & Paper Theory): (CO1 - 10M; CO2 - 10M; CO3-5M)	25
- CIE-3A: IA2 (Pen & Paper Theory): (CO3 - 5M; CO4 - 10M)	15
- CIE-3B: (Tutorial) (CO4 - 10M)	10
Semester End Exams (SEE):	
Mode of Exam	Pen and Paper (Theory)
Weightage of exam: - (CO1 - 5M; CO2 - 5M; CO3-15M; CO4-5M)	30

Course Outcomes- Programme Outcomes Matrix

Course Outcomes		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	Apply foundational concepts of functions, limits, continuity and derivatives to solve various engineering problems.	3	2			2			2	3			
CO2	Apply derivatives in problems including rate of change and single variable optimization.	3	2			2			2	3			
CO3	Apply integration in problems including calculation of areas and volumes.	3	2		2	2			2	3			
CO4	Apply gradient to implement gradient descent methods in optimization problems.	3	2		2	2			2	3			

Rubrics for evaluation of CIE (separate for each CIE component):

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Theory	Pen and paper written test will be conducted for 20 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Pen and paper written test will be conducted for 25 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Theory	Pen and paper written test will be conducted for 25 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty.

Total Marks		30
SEE Components		Rubrics for Assessment
SEE	Theory	Pen and paper written test will be conducted for 30 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty.